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Selective transport of plant root-associated bacterial populations in agricultural soils upon snowmelt

Dörte Dibbern [a], Andreas Schmalwasser [b], Tillmann Lueders [a,*], Kai Uwe Totsche [b]

[a] Institute of Groundwater Ecology, Helmholtz Zentrum München – German Research Center for Environmental Health, Ingolstädter Landstr. 1, 85764 Neuherberg, Germany
[b] Institute of Geosciences, Department of Hydrogeology, Friedrich-Schiller-Universität Jena, Burgweg 11, 07749 Jena, Germany

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ABSTRACT

Plants introduce abundant carbon into soils, where it is mineralised and sequestered. Proportions of this fresh organic carbon introduced to top soils can be relocated to deeper soil layers and even to groundwater by event-driven transport upon heavy rainfalls or after snowmelt. It is assumed that a significant fraction of this flux involves biocolloids and possibly microbial biomass itself. However, the nature of such transported microbes, their origin and the mechanisms of their mobilisation are still poorly understood. Here, we provide primary evidence that specific microbial populations are exported from top soils upon seepage events. At an experimental maize field, we have analysed the composition of mobilised bacterial communities collected in seepage water directly after snowmelt in winter at different depths (35 and 65 cm), and compared them to the corresponding bulk soil microbiota. Using T-RFLP fingerprinting and pyrotag sequencing, we reveal that mostly members of the *Betaproteobacteria* (*Methylophilaceae*, *Oxalobacteraceae*, *Comamonadaceae*), the *Alphaproteobacteria* (*Sphingomonadaceae*, *Bradyrhizobiaceae*), the *Gammaproteobacteria* (*Legionellaceae*) and the *Bacteroidetes* (*Sphingobacteriaceae*) were mobilised, all characteristic taxa for the rhizoplane. This highlights the importance of preferential flow along root channels for the vertical mobilisation and transport of microbes. Although the estimated quantitative fluxes of bacterial biomass carbon appeared low, our study allows for an improved understanding of the links between top soil, subsoil, and groundwater microbiota, as well as carbon fluxes between soil compartments.

1. Introduction

Soil organic matter (SOM) is the largest active carbon pool in terrestrial environments (Kindler et al., 2009) and it is known that carbon fluxes in soil can be considerable (Giardina et al., 2005). At the same time, many central mechanisms of carbon fluxes in soils are still poorly understood (Litton and Giardina, 2008). Quantity of SOM and carbon inputs is mostly determined by plants (Kögel-Knabner, 2002), however, SOM properties like stability, aggregation and reactivity are determined by the transforming organisms, most prominently soil microbes (Deckmyn et al., 2011; Kindler et al., 2009). Besides sequestration and mineralisation, transport of SOM by seepage water from top soils to deeper zones is an important factor contributing to carbon fluxes in soils (Kindler et al., 2011). These vertical fluxes represent a significant supply of fresh carbon to deeper soil layers and to the groundwater.

The mobile organic matter pool in soils not only comprises dissolved and colloidal organic carbon, but also biocolloids like bacteria, fungi and their fragments as well as viruses (Totsche et al., 2007). The translocation of colloids and particles, frequently along preferential flow paths including biopores can mediate fast and considerable mass transfer into deeper zones. It has been speculated that translocated microbes from the top soil could be an important source of biomass for subsoils, where they may significantly contribute to microbial activities (Jaesche et al., 2006).

The general understanding of the physical factors controlling vertical carbon transport through soil has improved over the last years (Bolan et al., 2011; Kalbitz and Kaiser, 2008). Already now, there is a basic grasp of bacterial transport mechanisms in soils, mostly focused on the transport of potential pathogens to groundwater (Natsch et al., 1996). Important factors inhibiting bacterial mobilisation are retention at air–water- and soil–water-interfaces, attachment and growth in biofilms, straining and also active adhesion (Sen, 2011). Soil bacteria can move actively in soils guided by chemotaxis (Sen, 2011) or may be mobilised and

* Corresponding author. Tel.: +49 89 3187 3687.
E-mail address: tillmann.lueders@helmholtz-muenchen.de (T. Lueders).

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transported passively by water flow (Unc and Goss, 2004), nematodes (Knox et al., 2004), growing roots (Feeney et al., 2006), or along fungal mycelia (Furuno et al., 2012). However, the highest fluxes of bacterial pathogens transported from upper to deeper soil layers occur after weather events producing abundant seepage water, such as long-lasting precipitation, flooding or snowmelt. Especially the detachment of top soil microbes by rain and snowmelt water with low-ionic strength is assumed to contribute to this mobilisation (Aislabie et al., 2011). Once mobilised, transport is assumed to be controlled mainly by the flow of seepage water along macropores, e.g. earth worm burrows, root channels and desiccation cracks (Natsch et al., 1996). Troxler and colleagues observed transport of amended bacteria from top soil down to depths of ~2.5 m only after heavy rainfalls (Troxler et al., 1998). The main route of transport was flow along macropores, which was confirmed also by other more recent studies (Bech et al., 2011; Jiang et al., 2010).

So, while many factors influencing the transport of carbon and bacteria over depth have been investigated, an understanding of the origin and nature of transported inherent soil bacteria, their contribution to carbon fluxes to deeper zones as well as their fate therein is currently lacking. Practically all studies on bacterial transport in soils and porous media have used only one or a few artificially amended bacterial species, and did not address mobilised natural bacterial communities. Therefore, here we investigated lysimeter samples at an experimental field site recently installed to unravel and quantify organismic and trophic carbon flow originating from maize plants (Kramer et al., 2012; Scharroba et al., 2012). We traced intrinsic bacterial communities mobilised by a snowmelt induced seepage event at different depths underneath the cropped field in late winter. Transported microbes were compared to bulk soil- and root-associated bacterial communities at corresponding depths. We hypothesise that 1) specific subsets of bacteria are selectively mobilised from the top soil and 2) the nature and composition of transported communities is controlled by the mechanisms of water flow. Furthermore, we postulate that 3) the transported bacteria contribute noticeably to net carbon flux. This work extends our current understanding of the event-driven mobilisation of microbes to deeper soils and groundwater, and has the potential to unravel translocation mechanisms of fresh plant-derived carbon inputs to deeper soils, as well as hidden links between depth-resolved microbial communities. Moreover, to the best of our knowledge, this represents the first study to systematically address the nature of a complex soil microbiota naturally mobilised by seepage water.

2. Materials and methods

2.1. The field site

Samples were taken from an experimental agricultural field located on a terrace plain of the river Leine, north–north-west of the City of Göttingen (Niedersachsen, Germany). The local climate, with a mean annual temperature of 8.7 °C and a mean annual precipitation of 645 mm, represents a temperate climate affected by the transgression from the maritime Atlantic climate on the west to the continental climate on the east. The elevation of the plane is 155–160 m a.s.l., striking towards north-west with a mean base slope of approximately 2%. According to IUSS (IUSS, 2007), the dominant soil types are Cambisols (Braunerden, KA5, 2005), Luvisols (Parabraunerden, KA5, 2005) and stagnic Luvisols (Pseudogley, KA5, 2005). Long agricultural use has severely affected the build-up of the soil profiles. The albic horizon typically found for these soils can no longer be detected in the field due to centuries of intensive tillage. In general, two plough layers (0.2 m and 0.3 m below surface) can be detected with particular strong compaction below the second plough layer: Bulk density (1.6 g cm^{-3}) in and below the second plough layer was relatively high (Supplementary Table S1). Most general chemical and physical characteristics of the soil have been previously reported (Kramer et al., 2012), but are listed here again in supplement together with some further relevant characteristics (Tables S1 & S2). In the agricultural year 2010, hybrid maize "Fernandez" (KWS Saat AG, Einbeck, Germany) was grown on the investigated field plot. This and also further relevant information on agricultural practices (fertilisation, pesticide treatments) can be found in Kramer et al. (2012).

To continuously collect seepage water, tension controlled lysimeters (KL2-100, UMS, Munich, Germany) were installed until the end of October 2009 directly below the plough horizon in approximately 35 cm depth and below the main rooted zone in 65 cm depth. The lysimeters were packed with undisturbed soil monoliths that were placed on top of a porous plate (pore size of 10 μm, SIC275, UMS, Munich, Germany). The porous plate served two needs: to allow for a collection of microbes and particles of up to a mean size of ~10 μm, and to enable the application of a matrix controlled suction for seepage water collection. The undisturbed soil monoliths were excavated at the locations of their latter re-installation at the same depth. Soil water potential was measured with a tensiometer (T8, UMS, Munich, Germany) installed at 35 cm depth. The respective suction was applied via a vacuum station (VS-twin, UMS, Munich, Germany).

2.2. Sampling

Before sampling, seepage water was collected at fortnightly intervals in 2 L glass bottles. In 2011, discharge into the subsoil was observed mainly during the fallow season in winter (December through March, 100 L m^{-2} at 35 cm depth and 86.5 L m^{-2} at 65 cm depth). Seepage water quantities below the plough horizon during the vegetation period from April through November 2011 were negligible (<0.1 L m^{-2}). Prior to actual sampling, an intensive snowmelt event in mid-January 2011 was followed by rain. Then, empty lysimeter collection bottles were installed within three successive time spans of 24 h (13, 14 and 17 January 2011), and fresh seepage water was sampled via the tension controlled lysimeters installed at 35 cm (L35) and 65 cm (L65) depth. Details on mean air and soil temperatures in the field as well as precipitation data during the period of investigation are given in the Supplementary Table S3 and Fig. S1. Organic carbon content in lysimeter water was analysed in samples from the first and third 24 h collection period, while bacterial community analysis was done with the second sample. All were successive samples, while the sampling procedure itself prevented true replicate water sampling. To a limited extent, as will be discussed further down, the two lysimeter samples obtained at distinct depth can be cautiously regarded as duplicate seepage water samples.

The sampling of lysimeter water always implies a certain contamination risk, as biofilms from the installation or from tubes can distort the composition of seepage water biota. We minimised this risk by using sterilised sampling bottles, inspecting the tubes for biofilms before the experiment, and mainly by maintaining a minimal retention time of fresh water samples in the lysimeters of only 24 h. Immediately after sampling, the water for the bacterial analyses was filtered (0.2 μm Corning, New York, USA), and filters and retained biomass were frozen at −20 °C until further processing.

At the same day, bulk soil samples were taken as composite samples of ten soil cores taken in a randomised manner via Pürkhauer coring across one 25 m × 25 m field plot as described previously (Kramer et al., 2012). Sampling was done at depths of 0–

10 cm (top soil, B10), 40–50 cm (plough layer, B50) and 60–70 cm (subsoil, B70) in the field plot containing the lysimeters. Furthermore, a root ball from a remaining maize stalk was also extracted. All soil and root samples were frozen within 6 h at −20 °C until further analyses. To obtain rhizosphere (Rh) and rhizoplane (Rp) samples, the root ball was thawed and then manually shaken until all readily detachable soil fell off. Root subsamples were then cut off with a sterile scalpel, and washed twice by shaking with 25 ml 1× PBS buffer in a 50 ml centrifuge tube (Buddrus-Schiemann et al., 2010). After washing, buffer and suspended solids were decanted into a fresh tube and the rhizosphere soil was collected by centrifugation (3345 rcf, 30 min). The remaining washed roots were chopped and used for rhizoplane nucleic acid extraction. All bulk soil, rhizosphere and rhizoplane samples were obtained in triplicates.

Seepage water analysis comprised among others, dissolved (DOC) and total organic carbon (TOC) (multi N/C 2100S, analytikjena, Germany). DOC samples were filtered with 0.45 μm syringe filters (Acrodisc 4508, Pall Corp., Port Washington, USA). Filtrate was then again filtered with 0.2 μm syringe filters (Acrodisc 4692, Pall Corp., Port Washington, USA) for better comparability with bacterial carbon.

2.3. DNA extraction

DNA from the filters, the soil samples (bulk and rhizosphere) and the root fragments was extracted with the same method. Filters and roots were cut into pieces (~2 cm) prior to extraction to fit into extraction tubes. Total DNA was extracted from the samples following a previously described procedure (Lueders et al., 2004) with some modifications: whole filters, ca. 0.4 g soil (ww) and ca. 0.3 g roots (ww) were used and bead beating was done in the presence of sodium phosphate, sodium dodecyl sulphate and phenol–chloroform–isoamylalcohol (25:24:1, pH 8). For purification and elimination of humic substances, silica gel columns were used from the DyeEx 2.0 Spin Kit (Qiagen GmbH, Hilden, Germany) for the DNA from soil and root samples.

2.4. T-RFLP fingerprinting

Bacterial community structure was analysed by 16S rRNA gene-targeted terminal restriction fragment length polymorphism (T-RFLP) fingerprinting with primers Ba27f-FAM/907r and subsequent *Msp*I digestion as described previously (Pilloni et al., 2011) with minor modifications (0.3 μM of each primer in the PCR reaction). Prior to the PCR the DNA from the rhizosphere samples and soil samples of 0–10 cm and 40–50 cm depth and were one hundred-fold diluted, DNA from rhizoplane and 60–70 cm depth were ten-fold diluted. DNA from the lysimeters remained undiluted. Bacterial T-RFLP data was processed with the T-REX online T-RF analysis software (Culman et al., 2008). Background noise filtering (Abdo et al., 2006) was on default factor 1 for peak heights and the clustering threshold for aligning peaks across the samples was set to 1 using the default alignment method (Smith et al., 2005). Relative T-RF abundance was inferred from peak heights.

2.5. Amplicon pyrosequencing

Amplicons from all samples were analysed by 454 pyrotag sequencing using bacterial primers (Ba27f/519r) as reported in detail previously (Pilloni et al., 2012). Briefly, pyrotag PCR was done under identical conditions as for fingerprinting, applying amplicon fusion primers with respective primer A or B adapters, key sequence and multiplex identifiers (MID) as reported (Pilloni et al., 2012). Amplicons were purified and pooled in equimolar 10^9 μl^{-1} concentration. Emulsion PCR, emulsion breaking and sequencing were performed as previously described in detail (Pilloni et al., 2012) following manufacturer protocols using a 454 GS FLX pyrosequencer and Titanium chemistry (Roche Applied Biosystems, Penzberg, Germany). Demultiplexed, trimmed and quality-filtered reads >250 bp (Pilloni et al., 2012) were classified using the RDP pyrosequencing pipeline (Wang et al., 2007). A summary of raw and trimmed read statistics for the different pyrotag libraries generated in this study is available in supplement (Table S4). For graphical plotting of taxa abundances, all taxa containing less than four reads between all samples were excluded and read numbers were converted to relative read abundances.

Bidirectional amplicon pools of matching sequences were assembled into contigs with the SEQMAN II software (DNAStar) using assembly thresholds of at least 97% sequence similarity over a 50 bp match window for T-RF prediction (Pilloni et al., 2012). Only contigs containing at least one forward and one reverse read were used to predict *in-silico*-T-RFs for dominating consensus phylotypes using TRiFLe (Junier et al., 2008) and thus linking of T-RF and pyrotag data (Pilloni et al., 2012; Weissbrodt et al., 2012). Resulting sequencing data have been deposited with the NCBI sequence read archive (SRA accession numbers: SRX 361197, SRX 363682-86).

2.6. Statistical analysis

The abundance data sets from both the T-RFLP fingerprinting and amplicon sequencing were arcsin transformed (arcsin$\sqrt{}$) and interpreted by principal component analysis (PCA) to depict the similarities of the bacterial communities in the different samples. The function 'PCA' was used from the package FactoMineR (Le et al., 2008) from the open source statistic program R (R Development Core Team, 2011).

2.7. Quantitative PCR

Quantification of 16S RNA gene abundance in extracts was done by quantitative PCR as described previously (Winderl et al., 2008) with minor modifications. Genes per extract were quantified in three dilutions with triplicate pipetting each. Appropriate dilutions were 10^{-3}, 10^{-4} and 10^{-5} for the soil of 0–10 cm, 50–70 cm and the rhizosphere; 10^{-2}, 10^{-3} and 10^{-4} for 60–70 cm and rhizoplane extracts, as well as 10^{-1}, 10^{-2} and 10^{-3} for the lysimeter water samples.

2.8. Inference of bacterial biomass carbon

Two estimation approaches were used to infer bacterial biomass carbon in lysimeter samples by the means of the qPCR data. (A) In the first approach, we determined pyrotag read abundances for the most important phylogenetic lineages representing over 90% of all sequence reads in each sample. We have previously shown that taxon read abundances generated by our pyrotag pipeline are reproducible and semi-quantitatively robust (Pilloni et al., 2012). Then, the average *rrn* operon copy number per cell for these families and lineages was searched in the IMG database (Markowitz et al., 2012). Subsequently, an extensive literature search was done for a reasonable estimate of the average carbon content per bacterial cell in agricultural soils. Unfortunately, most respective studies refer to marine bacteria and pure or enrichment cultures. We found a wide range of literature values between 1.17–214 fg carbon per cell (Bratbak, 1985; Fagerbakke et al., 1996; Loferer-Krossbacher et al., 1998; Tuomi et al., 1995). We decided to adapt values from Troussellier et al. (1997), who used cultures of five marine and five non-marine species under starving conditions and reported an average of 26.42 ± 1.08 fg carbon per cell, which appeared an adequate range. (B) As control, we consulted data on microbial biomass carbon obtained by chloroform-fumigation from the same soil samples (Kramer et al., 2012) and correlated those with 16S rRNA gene qPCR counts from the same samples and depths. However, since only data for total microbial biomass carbon was available (including prokaryotes and microeukaryotes), we estimated the fungi: bacteria biomass ratio by following Joergensen and Wichern (2008), who defined a general bacterial contribution to total microbial carbon of 40%–85% in agricultural soils.

2.9. SEM images from seepage water samples

The morphology of colloids and particles in the seepage water was investigated with scanning electron microscopy (SEM). Prior to SEM imaging, seepage water samples (10 μl) were transferred to a silicon wafer (PLANO GmbH, Wetzlar, Germany) and air-dried. High resolution images from seepage water samples were taken with an Ultra PLUS field emission scanning electron microscope (Carl Zeiss, Oberkochen, Germany), operating with 2 kV acceleration voltage.

3. Results

To investigate natural bacterial communities mobilised and transported by seepage water in an agricultural soil, we sampled fresh lysimeter water over 24 h after an event of snowmelt and rain in January 2011. This was the minimal time span needed to obtain sufficient seepage water (50 and 75 ml) at the depths of 38 cm (L35) and 65 cm (L65) respectively, where lysimeters were installed. In this short time span 'bottle effects' of sampled seepage water were probably avoided. Corresponding bulk soil near the lysimeters was sampled by coring at 0–10 cm (B10), 40–50 cm (B50) and 60–70 cm (B70) on the same day. Additionally, rhizosphere (Rh) and rhizoplane samples (Rp) from a remaining capped maize stalk were also sampled. Due to the very limited amount of available seepage water, replicate sampling was not possible, and technical PCR replication only was possible for lysimeter DNA extracts. For soil and root DNA extracts, biological replication was applied, of course.

3.1. Bacterial community patterns in water and soil

T-RFLP and sequencing data sets were first screened with principal component analysis (PCA) to reduce complexity and allow overall comparisons. For both analysing methods, PCA arranged the samples in a very similar pattern (Fig. 1). Lysimeter samples grouped together and were discriminated from the bulk soil communities by a negative PC1, together with the rhizoplane samples. PC1 accounted for 37% of overall T-RF and 60% of overall pyrotag abundance variability. Rhizosphere samples were placed closest to the 10 cm bulk soil samples in the PCA biplots, but with a tendency towards rhizoplane samples. Potentially, a clearer distinction between rhizosphere and bulk soil samples may have been prevented by our sampling procedure, where the freezing and thawing of entire root balls preceded compartment separation.

Nevertheless, while PC1 seemed mostly influenced by the sampled compartment, PC2 appeared predominantly controlled by depth. In general, triplicate fingerprints from technical (water) or biological (soil, roots) replicates were well comparable as visualised by PCA of T-RFLP fingerprints (Fig. 1A). Only one replicate of the 40–50 cm bulk soil DNA extracts appeared more similar to the 0–10 cm bulk soil samples than the other two replicates.

PCA biplots also depicted T-RFs and bacterial taxa identified in pyrotag sequencing as vectors with their impact on sample ordination. The bacterial taxa (Fig. 1B) representative for lysimeter water were mostly within the *Betaproteobacteria* (e.g. *Methylophilaceae*, *Oxalobacteraceae* and *Comamonadaceae*), but also within the *Alphaproteobacteria* (e.g. *Sphingomonadaceae* and *Bradyrhizobiaceae*), *Gammaproteobacteria* (e.g. *Legionellaceae*) and *Bacteroidetes* (e.g. *Sphingobacteriaceae*). In contrast, reads affiliated to *Acidobacteria* (e.g. the subgroups 1, 4, 6, and 7) (Jones et al., 2009), *Firmicutes* (e.g. *Bacillaceae*, *Paenibacillaceae*) and *Actinobacteria* (e.g. *Nocardiaceae* and *Micromonosporaceae*) were clearly more abundant in bulk soil compartments (Fig. 1B).

To further support linking of both T-RF and pyrotag data sets, we aimed to match some of the identified discriminative T-RFs to assembled dominating amplicon contigs from the same samples (Pilloni et al., 2012). Although not unambiguous for all relevant T-RFs, consistencies between both approaches were revealed. Hence, we could observe that a majority of the *Bradyrhizobiaceae*- and *Sphingomonadaceae*-related reads were assigned to the T-RF of 149 bp and most members of the *Comamonadaceae* and *Moraxellaceae* comprised a fragment of 484 bp. *Legionellaceae* could be assigned to T-RFs of 490 bp and 497 bp and *Methylophilaceae*, *Oxalobacteraceae* and *Pseudomonadaceae* to a fragment of 490 bp. All of these T-RFs

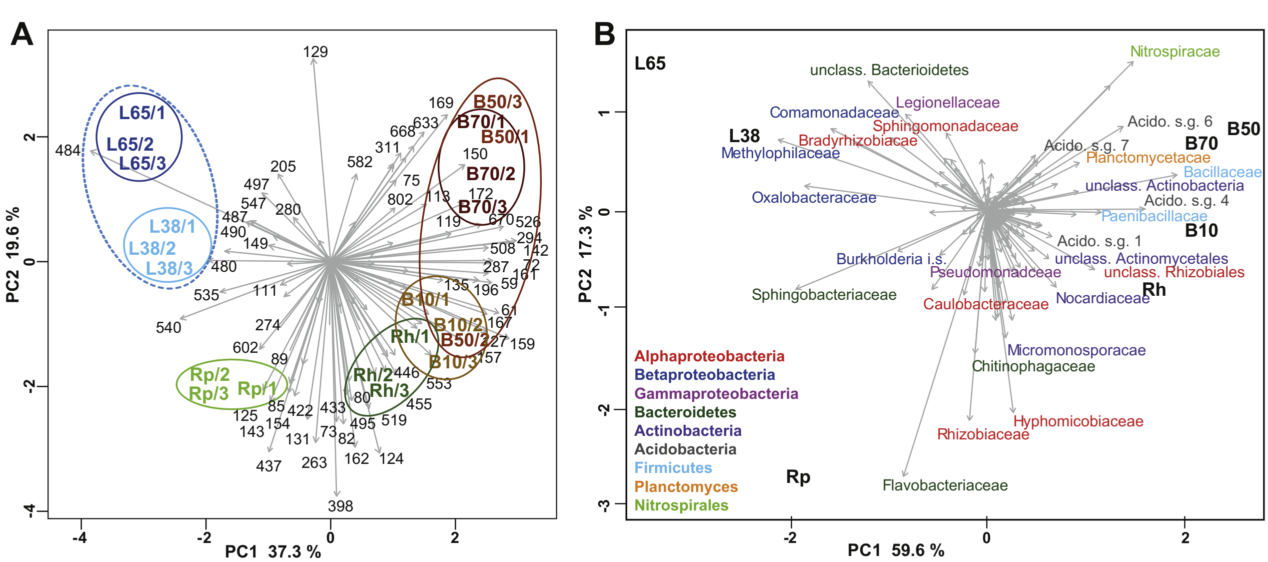

Fig. 1. PCA biplots of T-RFLP (A) and amplicon sequencing (B) data sets. Sample codes are given in bold, selected relevant T-RFs or taxa in non-bold. T-RFs (A) and taxa (B) are denoted by grey arrows, but only selected discriminative taxa are designated. Variance explanation proportions for each principal component (PC) are given. L38, L65: lysimeter samples from 38 cm to 65 cm depth; B10, B50, B70: bulk soil samples from 0–10, 40–50 and 60–70 cm depth; Rh: rhizosphere; Rp: rhizoplane.

and lineages were clearly characteristic for lysimeter samples. Moreover, reads within the *Chitinophagaceae*, typical for rhizoplane samples, were matched to T-RFs of 85 bp and 89 bp and both T-RFs tended towards rhizoplane samples in the PCA biplot. Also, several typical bulk soil taxa were assigned to T-RFs frequent in the respective samples: *Bacillaceae*: 150 bp; *Paenibacillaceae*: 161 bp, *Acidobacteria*: subgroup 4: 142 bp; subgroup 6: 294 bp; subgroup 7: 142 bp; *Nocardiaceae*: 135 bp and *Micromonosporaceae*: 159bp (Fig. 1).

3.2. Differential mobilisation efficiency of bacterial lineages

As bacteria from the lysimeter samples were clearly not a representative subset of total soil bacterial communities, selective mobilisation and transport were suggested. In fact, relative sequence abundances of overall phyla differed profoundly between lysimeter water and soil samples (Fig. 2). As mentioned above, also the taxa most prominently mobilised by seepage water were not likewise abundant in bulk soil samples. Most prominently, Gram-positive bacteria like *Firmicutes* and *Actinobacteria* were abundant in bulk soil and rhizosphere (8–16% read abundance), but almost absent in Lysimeter samples (below 1%, Fig. 2). Instead, many taxa abundant in lysimeter water were also frequent on the rhizoplane, i.e. the *Oxalobacteraceae* (L35: 10%, L65: 12%, Rp: 8%), *Comamonadaceae* (L35: 4%, L65: 17%, Rp: 7%) and *Sphingobacteriaceae* (L35: 7%, L65: 6%, Rp: 7%). For other lineages, however, relative sequence abundances were low for all soil and root samples, and relatively high in lysimeter water: i.e. the *Opitutaceae* (L35: 3%), *Bradyrhizobiaceae* (L35: 2%, L65: 13%), *Sphingomonadaceae* (L35: 4%, L65: 7%), *Methylophilaceaea* (L35: 6%, L65: 10%) and *Legionallaceae* (L35: 3%, L65: 2%). Finally, different families appeared distinctly abundant in seepage water at the two depths. I.e. reads of the *Gammaproteobacteria* were much more abundant at 35 cm (20%) than at 65 cm (7%), whereas *Alpha*- and *Betaproteobacteria* were more frequent in lysimeter water at 65 cm (*Alpha*.: 23%, *Beta*.: 45%) than at 35 cm depth (*Alpha*.: 11%, *Beta*.: 24%).

3.3. Bacterial contribution to mobilised organic carbon

To verify that intact bacterial cells and not just cell fragments or free DNA were transported into the lysimeter flasks, scanning electron microscopy was used. Indeed, many visualised structures resembled undamaged bacterial cells (Fig. 3). Therefore, we concluded that at least a noticeable amount of intact bacteria was mobilised by seepage water. Total organic carbon (TOC) and dissolved organic carbon (DOC) were measured in seepage water and particulate organic carbon (POC) was calculated from those values (Table 1). Overall, we observed a TOC transport by seepage water of 1.1 g C m^{-2} below the plough horizon and 0.8 g C m^{-2} below the main root zone in the hydraulic year 2011. The fraction of POC was 6% (0.07 g C m^{-2}) for 35 cm and 14% (0.10 g C m^{-2}) for 65 cm depth.

The contribution of bacterial biomass carbon to this efflux was estimated based on 16S RNA gene abundances quantified in seepage water by quantitative PCR (Table 2). Varying 16S RNA gene copy numbers for different bacterial species were taken into account. Average 16S rRNA gene operon copy numbers for important bacterial lineages of each sample (Suppl. Table S5), deduced from the integrated microbial genome database (IMG; Markowitz et al., 2012) were weighted according to their abundances resulting in a weighted mean operon copy number for every sample. Thus, for lysimeter waters, an average of 2.69 (L35) and 2.47 (L65) copies per cell were deduced (Table 2).

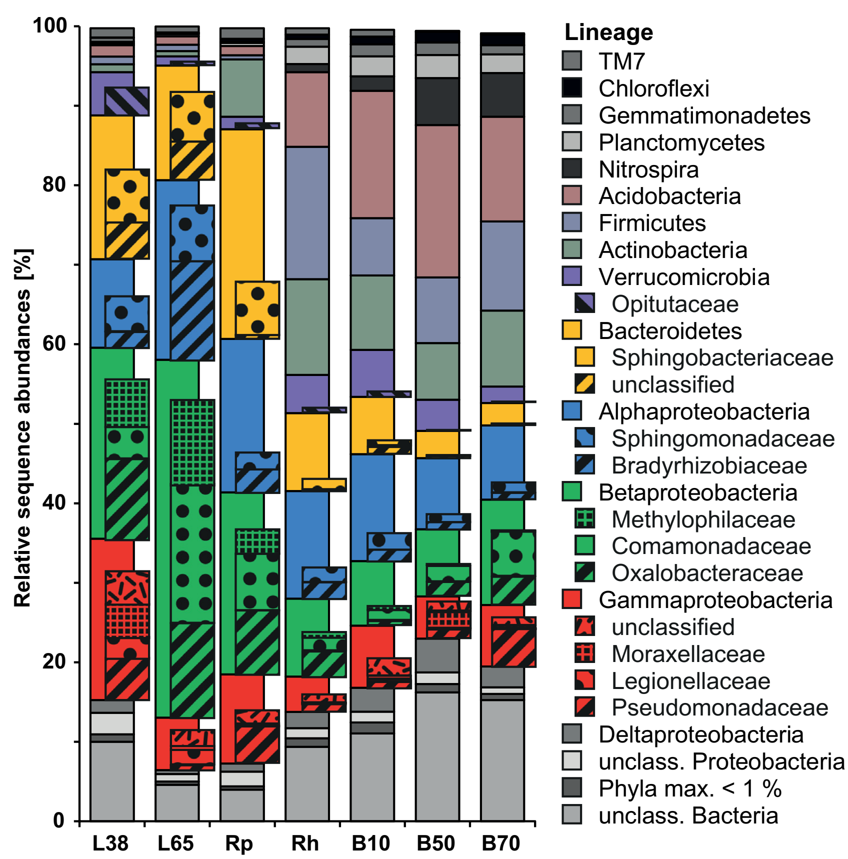

Fig. 2. Relative sequence abundance for overall taxa obtained in amplicon sequencing. Selected highly discriminative sub-phylum taxa are highlighted. Sample codes are as in Fig. 1.

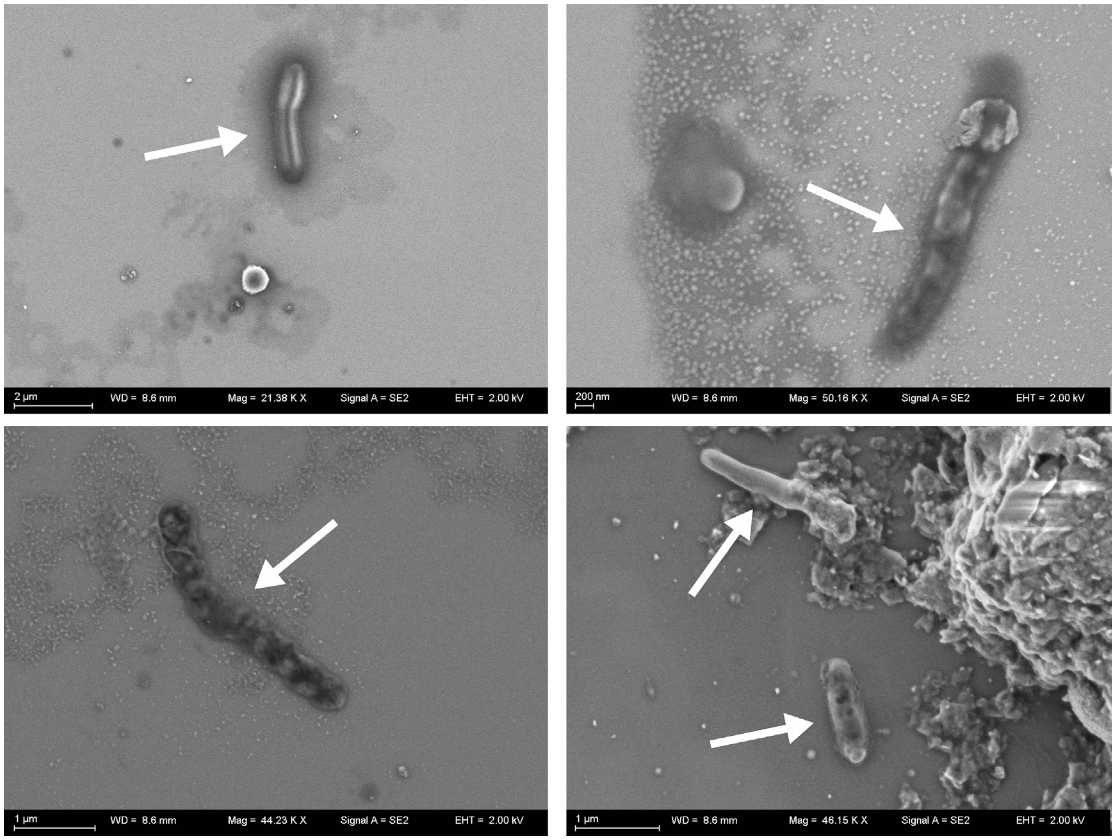

Fig. 3. Representative scanning electron images of seepage water samples collected in lysimeters after the snowmelt event in January 2011. Presumed intact bacterial cells are indicated by arrows.

From this, together with estimated cellular abundances, mobilised bacterial carbon per water sample was inferred with an assumption of 26.27 fg carbon per cell (Troussellier et al., 1997). We also compared our calculations for bacterial carbon from soil samples with the measured microbial biomass carbon from the same soils (Kramer et al., 2012). Considering that between 40% and 85% of total microbial biomass carbon is considered to be of bacterial origin, both estimates were in the same order of magnitude (Table 1). According to these estimates, only 0.01%–0.02% of total organic carbon mobilised in seepage water upon snowmelt related seepage events originated from actual bacterial cells, a ratio appearing surprisingly low (Table 2) upon first thought.

4. Discussion

Soil bacterial community composition is known to differ significantly with depth due to changing habitat parameters like carbon and nitrogen supply, pH, and moisture (Cruz-Martinez et al.,

Table 1
Estimation of bacterial biomass carbon transported in seepage water.

Sample[a]	L38 (u = ml)	L65 (u = ml)	B10 (u = g)	B50 (u = g)	B70 (u = g)
16s RNA gene abundance					
qPCR [copies u^{-1}]	$1.20 \times 10^5 \pm 5.04 \times 10^4$	$5.50 \times 10^4 \pm 1.75 \times 10^4$	$5.50 \times 10^9 \pm 1.99 \times 10^9$	$8.54 \times 10^8 \pm 3.34 \times 10^8$	$6.52 \times 10^8 \pm 2.91 \times 10^8$
Average *rrn* copies per cell					
Weighted mean of families[b]	2.69	2.47	2.67	2.67	2.96
Bacterial cell number					
Copies/copies per cell [cells u^{-1}]	$4.48 \times 10^4 \pm 1.87 \times 10^4$	$2.23 \times 10^4 \pm 7.10 \times 10^3$	$2.06 \times 10^9 \pm 7.44 \times 10^8$	$3.20 \times 10^8 \pm 1.25 \times 10^8$	$2.20 \times 10^7 \pm 9.84 \times 10^7$
Bacterial biomass carbon					
Cell number × 26.27 fg carbon $cell^{-1}$ [μg Cu^{-1}]	$1.18 \times 10^{-3} \pm 4.92 \times 10^{-4}$	$5.85 \times 10^{-4} \pm 1.86 \times 10^{-4}$	$5.41 \times 10^1 \pm 1.96 \times 10^1$	8.4 ± 3.28	5.78 ± 2.58
Microbial biomass carbon					
Measured for soil samples [μg Cu^{-1}]	–	–	1.24×10^2[c]	1.11×10^1[c]	9.86[c]
Bacterial biomass carbon					
40–85% of microbial carbon [μg Cu^{-1}]	–	–	4.96×10^1–1.06×10^2	4.42–9.4	3.94–8.38

[a] Abbreviations are as explained in the legend of Fig. 1.
[b] Taken from the IMG database (Markowitz et al., 2012).
[c] Taken from Kramer et al. (2012).

Table 2
Organic carbon pools and estimated contribution of bacterial biomass carbon.

Sample[a]	L38 (u = ml)	L65 (u = ml)	B10 (u = g soil)	B50 (u = g soil)	B70 (u = g soil)
TOC (OC > 0.45 μm) [μg u^{-1}]	5.13 ± 0.27	5.97 ± 0.13	11.3×10^3	10.3×10^3	4.4×10^3
DOC (OC < 0.45 μm) [μg u^{-1}]	4.77 ± 0.08	5.53 ± 0.15	–	–	–
OC < 0.2 μm [μg u^{-1}]	2.87 ± 0.74	3.67 ± 0.54	–	–	–
Bacterial contribution to TOC [%]	0.02	0.01	0.48	0.08	0.13
Bacterial contribution to DOC [%]	0.02	0.01	–	–	–
Bacterial contribution to OC > 0.2 μm [%]	0.04	0.02	–	–	–

[a] Abbreviations are as explained in the legend of Fig. 1.

2012; Gelsomino and Azzellino, 2011; Will et al., 2010). Although bacteria from deeper soil layers seem well adapted to their environment, they are discussed to largely originate from top soil initially (Holden and Fierer, 2005; Kieft et al., 1998). The prime mechanism of vertical transport into subsoils is advective transport with seepage water (Balkwill et al., 1998). Here, we examined whether bacteria from natural soil microbial communities are mobilised selectively, and we discuss the mechanisms which could control their transport by seepage water, at least for the investigated sampling time point in winter.

Fresh (24 h) lysimeter water from two depths, soil from the same depths as well as top soil, rhizosphere and the rhizoplane of a remaining root stalk were sampled. This represents a unique set of corresponding soil and seepage water samples never investigated before to the best of our knowledge. However, it relates to only one time point of sampling during late winter, were the biological activity in soil is low. Therefore, our study provides only a first insight into possible mechanisms of selective vertical bacterial mobilisation under natural conditions, with a focus on an agricultural soil and seepage water collected in the fallow season.

As mentioned, sampling of replicate lysimeter water samples was technically not feasible due to the time span (~24 h) required to obtain sufficient water. A water sample taken on the next day would also not have been a replicate, but a successive sample. Still, our study reports on two parallel lysimeter waters collected at different depths. Within certain limitations, these can cautiously also be regarded as replicates. And indeed, we saw very similar transported microbes and consistent distinctions to (replicated) bulk soil microbiota in these parallel samples. Therefore, we are confident that our data interpretations and main conclusions are robust, even without proper replication of water samples. A much more comprehensive study with seasonal sampling, seepage water replicates, as well as bacterial amendments and artificial precipitation is currently still in progress. Nevertheless, we believe that already the initial observations reported here can be seminal to improve our understanding of the vertical transport of natural microbial populations in soil.

4.1. Selective mobilisation and transport of bacteria

In our study, bacterial communities transported by seepage water were distinct especially from bulk soil communities, indicating that transported bacteria were not merely a subset of the total soil microbiota. Rather, bacteria appeared to be mobilised selectively, thus confirming our initial hypothesis. Taxa within the *Bacteroidetes*, *Beta*- and *Alphaproteobacteria*, especially abundant on the rhizoplane, were transported preferentially, in contrast to *Actinobacteria*, *Firmicutes* and *Acidobacteria*. Bacteria from those phyla were hardly detected in lysimeter water, although abundant in bulk soil. Despite intensive literature research, no other study was found directly demonstrating such a selective mobilisation of natural microbial communities in soil.

In contrast, mobilisation and transport of pathogens from manure or artificial amendments have been intensively investigated (for a comprehensive review see Bradford et al., 2013). Mechanisms of transport observed here should also be valid for native soil bacteria. Mobilisation of bacteria amended to the top soil in unsaturated media is generally impeded by straining because of pore size filtering and interactions with soil matrix (Sen, 2011). Therefore, bacteria are mostly transported by sporadic events with excessive seepage water like heavy rain or snowmelt (Jiang et al., 2010; Natsch et al., 1996). Then, quasi-saturated conditions prevail and water flux is dominated by preferential flow capable of flushing bacteria through macropores, i.e. wormholes, root channels and cracks (Bech et al., 2011; Unc and Goss, 2004). These mechanisms can be assumed to be even more relevant at our field site considering the highly condensed plough layers.

The bacterial community from rhizoplane samples was clearly most similar to the lysimeter microbiota. Therefore, we conclude that a significant amount of transported bacteria may actually have been mobilised by water flow along root channels. In disturbed soils like our agricultural field, root channels can form the majority of macropores (Ghestem et al., 2011). Roots shrink and decay in winter and give even more space for water flow. Besides being the main flow route of seepage water, root channels harbour high amounts of bacteria in comparison to the surrounding soil and are known to be ‘hot spots’ of bacterial activity (Bundt et al., 2001; Vinther et al., 1999). Such macropores feature relatively high nutrient supply from decaying roots and washed out carbon compounds. One may speculate that microbial communities in the bulk soil could be better protected against mobilisation by seepage water, as they often reside within micropores or are attached to soil particles.

Yet, bacterial taxa from lysimeter samples were not of totally equal abundance as in rhizoplane samples. Therefore also other mechanisms may influence bacterial mobilisation. For example *Actinobacteria*, although abundant on the rhizoplane, were not noticeably transported, which could be ascribed to the branched mycelia many *Actinobacteria* form in their active state (Balkwill et al., 1998). Spores of *Actinobacteria*, however, have been found in high abundances in deeper soil layers under high recharge, indicating elevated transport by seepage water (Balkwill et al., 1998). Generally, hydrophobicity, surface charge, size and cell structure are features known to influence mobilisation rates of bacteria (Bradford et al., 2013; Gargiulo et al., 2008). How these cell properties specifically affected mobilisation in our study cannot be predicted, as these characteristics are not only species-specific, but also vary with growth state and physical condition (Foppen and Schijven, 2006).

Nevertheless, we observed that some bacteria were more readily mobilised than others. Although quite rare in top soil, rhizosphere and rhizoplane biota, some bacteria were quite abundant in lysimeter samples, e.g. populations within the unclassified *Bacteroidetes*, *Opitutaceae*, *Legionellaceae* and *Moraxellaceae*. For *Legionella* spp., a life cycle with sessile biofilm and mobile swarming cells is known (Declerck, 2010), and boosts of *Legionella* infections have been related to heavy rainfalls (Fisman et al., 2005; Hicks et al., 2007). This and their relatively high abundance in lysimeter samples indicate that for some bacteria including potential pathogens, transport by seepage water could even be a key distribution strategy.

Our conclusions are largely based on relative T-RF and sequence read abundances. Since pyrotag sequencing is a relatively new method, its semi-quantitative accuracy is still discussed. Although some reports are quite supportive (Lee et al., 2012; Pilloni et al., 2012), it is clear that quantitative assertions inferred from such data sets must be interpreted with caution (Amend et al., 2010), especially, if samples are not analysed in biological replicates due to the still considerable sequencing costs. Therefore, we also used T-RFLP fingerprinting with three biological (soil samples) or technical (water samples) replicates to verify the most important findings from amplicon sequencing. T-RFLP fingerprinting is known to allow for a robust semi-quantitative evaluation of OTU abundances between samples treated in the same way (Schütte et al., 2008).

We show that bacterial community distinctions via fingerprinting and amplicon sequencing yielded very similar results in PCA biplots, and that many important bacterial taxa, discriminative for a given compartment at this time point of sampling, were assigned to defined T-RFs of similar discriminative character, which strengthens our overall interpretation of the pyrotag data.

4.2. Bacterial contribution to vertical carbon flux

We are aware of the many assumptions we used in the estimation of bacterial cellular carbon content in seepage water. Yet, data handling was chosen carefully and the consistent results between the two estimation approaches confirms at least the correct magnitude of our resulting interpretation. More direct measures like cell counting or chloroform-fumigation-extraction (Daims et al., 2001; Vance et al., 1987) were not feasible in our study because of the small amount of water from the lysimeters and low cell densities therein. Despite large amounts of seepage water formed upon the event, only a fraction of the total precipitation (42% in 35 cm; 6% in 65 cm) reached the lysimeters. This may be caused by (a) evapotranspiration, (b) bypassing of lysimeters by macropore flow, (c) lateral interflow on top of the compacted plough-layer and (d) to a lesser extent, saturation of the lysimeter and the potential bypassing as a consequence of excess water saturation. Nevertheless, 42% collection of the total precipitation at 35 cm depth is in good agreement with measurements from comparable filed sites.

Organic carbon content in seepage water was measured from lysimeter samples prior and after sampling of bacteria, but within the same weather event of rain and snowmelt. As carbon content was very reproducible in those samples (Table 2), we confidently assume the same amount of carbon in the samples we analysed the bacterial community from. In contrary to our hypothesis, bacterial biomass appeared to contribute only marginally to carbon transported to deeper soil zones. We initially assumed much higher proportions of bacterial carbon because of the relevance of colloidal carbon reported in other studies (Gao et al., 2006; Martins et al., 2013; Totsche et al., 2007) and the overlapping size fractions between colloids (here: 0.35–1 μm) and soil bacteria (ca. 0.2–1 μm). Partly, this may be attributed to the fact that we collected our samples after several days of snowmelt and subsequent rain. Potentially, the biggest load of bacterial cells mobilisable by the event had already been washed out. It has been reported that amounts of mobilised organic carbon and colloids are highest after long dry weather conditions followed by excessive rain and seepage water (Totsche et al., 2007). So, seasonal and also time-resolved contribution of bacterial biomass to vertical carbon transport remains to be elucidated.

Still, bacteria remain at least indirectly important for carbon transport into deeper soil layers. Microbes turn over 85–90% of soil organic carbon (Ekschmitt et al., 2008) and up to 80% of SOM can be derived from dead microbial biomass (Liang and Balser, 2011). Therefore, bacteria and fungi determine the composition of soil organic carbon and all of its properties relevant for mobilisation, including size, hydrophobicity and charge. Especially fragments from cell walls, proteins and peptides interact strongly with soil particles (Fan et al., 2004; Kögel-Knabner, 2002; Miltner et al., 2012) implicating enhanced immobilisation of those compounds.

4.3. Conclusions

To the best of our knowledge, this is the first report to systematically address the mobilisation and transport of natural soil bacterial communities in seepage water. As hypothesised, specific bacteria were mobilised selectively, likely due to preferential flow of seepage water along root channels. Still, compared with total transported organic carbon, the contribution of mobilised bacterial biomass to carbon fluxes between compartments appeared small. Nevertheless, this study allows for an interesting new perspective of potential mechanisms linking top and subsoil microbial communities, as well as of the origin and diversity of subsoil and even groundwater microbiota including potential pathogens.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.soilbio.2013.10.040.

References

Abdo, Z., Schuette, U.M.E., Bent, S.J., Williams, C.J., Forney, L.J., Joyce, P., 2006. Statistical methods for characterizing diversity of microbial communities by analysis of terminal restriction fragment length polymorphisms of 16S rRNA genes. Environ. Microbiol. 8, 929–938.

Aislabie, J., McLeod, M., Ryburn, J., McGill, A., Thornburrow, D., 2011. Soil type influences the leaching of microbial indicators under natural rainfall following application of dairy shed effluent. Soil Res. 49, 270–279.

Amend, A.S., Seifert, K.A., Bruns, T.D., 2010. Quantifying microbial communities with 454 pyrosequencing: does read abundance count? Mol. Ecol. 19, 5555–5565.

Balkwill, D.L., Murphy, E.M., Fair, D.M., Ringelberg, D.B., White, D.C., 1998. Microbial communities in high and low recharge environments: implications for microbial transport in the vadose zone. Microb. Ecol. 35, 156–171.

Bech, T., Dalsgaard, A., Jacobsen, O.S., Jacobsen, C.S., 2011. Leaching of *Salmonella enterica* in clay columns comparing two manure application methods. Ground Water 49, 32–42.

Bolan, N.S., Adriano, D.C., Kunhikrishnan, A., James, T., McDowell, R., Senesi, N., 2011. Dissolved organic matter: biochemistry, dynamics and environmental significance in soils. In: Sparks, D.L. (Ed.), Advances in Agronomy, vol. 110. Elsevier Academic Press Inc., San Diego, pp. 1–75.

Bradford, S.A., Morales, V.L., Zhang, W., Harvey, R.W., Packman, A.I., Mohanram, A., Welty, C., 2013. Transport and fate of microbial pathogens in agricultural settings. Crit. Rev. Environ. Sci. Technol. 43, 775–893.

Bratbak, G., 1985. Bacterial biovolume and biomass estimations. Appl. Environ. Microbiol. 49, 1488–1493.

Buddrus-Schiemann, K., Schmid, M., Schreiner, K., Welzl, G., Hartmann, A., 2010. Root colonization by Pseudomonas sp. DSMZ 13134 and impact on the indigenous rhizosphere bacterial community of barley. Microb. Ecol. 60, 381–393.

Bundt, M., Widmer, F., Pesaro, M., Zeyer, J., Blaser, P., 2001. Preferential flow paths: biological 'hot spots' in soils. Soil Biol. Biochem. 33, 729–738.

Cruz-Martinez, K., Rosling, A., Zhang, Y., Song, M.Z., Andersen, G.L., Banfield, J.F., 2012. Effect of rainfall-induced soil geochemistry dynamics on grassland soil microbial communities. Appl. Environ. Microbiol. 78, 7587–7595.

Culman, S.W., Gauch, H.G., Blackwood, C.B., Thies, J.E., 2008. Analysis of T-RFLP data using analysis of variance and ordination methods: a comparative study. J. Microbiol. Methods 75, 55–63.

Daims, H., Ramsing, N.B., Schleifer, K.-H., Wagner, M., 2001. Cultivation-independent, semiautomatic determination of absolute bacterial cell numbers in environmental samples by fluorescence in situ hybridization. Appl. Environ. Microbiol. 67, 5810–5818.

Deckmyn, G., Campioli, M., Muys, B., Kraigher, H., 2011. Simulating C cycles in forest soils: including the active role of micro-organisms in the ANAFORE forest model. Ecol. Model. 222, 1972–1985.

Declerck, P., 2010. Biofilms: the environmental playground of *Legionella pneumophila*. Environ. Microbiol. 12, 557–566.

Ekschmitt, K., Kandeler, E., Poll, C., Brune, A., Buscot, F., Friedrich, M., Gleixner, G., Hartmann, A., Kastner, M., Marhan, S., Miltner, A., Scheu, S., Wolters, V., 2008. Soil-carbon preservation through habitat constraints and biological limitations on decomposer activity. J. Plant Nutr. Soil Sci.-Z. Pflanzenernahr. Bodenkd. 171, 27–35.

Fagerbakke, K.M., Heldal, M., Norland, S., 1996. Content of carbon, nitrogen, oxygen, sulfur and phosphorus in native aquatic and cultured bacteria. Aquat. Microb. Ecol. 10, 15–27.

Fan, T.W.M., Lane, A.N., Chekmenev, E., Wittebort, R.J., Higashi, R.M., 2004. Synthesis and physico-chemical properties of peptides in soil humic substances. J. Pept. Res. 63, 253–264.

Feeney, D.S., Crawford, J.W., Daniell, T., Hallett, P.D., Nunan, N., Ritz, K., Rivers, M., Young, I.M., 2006. Three-dimensional microorganization of the soil-root-microbe system. Microb. Ecol. 52, 151–158.

Fisman, D.N., Lim, S., Wellenius, G.A., Johnson, C., Britz, P., Gaskins, M., Maher, J., Mittleman, M.A., Spain, C.V., Haas, C.N., Newbern, C., 2005. It's not the heat, it's the humidity: wet weather increases Legionellosis risk in the greater Philadelphia metropolitan area. J. Infect. Dis. 192, 2066–2073.

Foppen, J.W.A., Schijven, J.F., 2006. Evaluation of data from the literature on the transport and survival of *Escherichia coli* and thermotolerant coliforms in aquifers under saturated conditions. Water Res. 40, 401–426.

Furuno, S., Remer, R., Chatzinotas, A., Harms, H., Wick, L.Y., 2012. Use of mycelia as paths for the isolation of contaminant-degrading bacteria from soil. Microb. Biotechnol. 5, 142–148.

Gao, B., Saiers, J.E., Ryan, J., 2006. Pore-scale mechanisms of colloid deposition and mobilization during steady and transient flow through unsaturated granular media. Water Resour. Res. 42, W01410.

Gargiulo, G., Bradford, S.A., Simunek, J., Ustohal, P., Vereecken, H., Klumpp, E., 2008. Bacteria transport and deposition under unsaturated flow conditions: the role of water content and bacteria surface hydrophobicity. Vadose Zone J. 7, 406–419.

Gelsomino, A., Azzellino, A., 2011. Multivariate analysis of soils: microbial biomass, metabolic activity, and bacterial-community structure and their relationships with soil depth and type. J. Plant Nutr. Soil Sci. 174, 381–394.

Ghestem, M., Sidle, R.C., Stokes, A., 2011. The influence of plant root systems on subsurface flow: implications for slope stability. Bioscience 61, 869–879.

Giardina, C.P., Coleman, M.D., Binkley, D., Hancock, J.E., King, J.S., Lilleskov, E.A., Loya, W.M., Pregitzer, K.S., Ryan, M.G., Trettin, C.C., 2005. The Response of Belowground Carbon Allocation in Forests to Global Change. In: Tree Species Effects on Soils: Implications for Global Change, pp. 119–154.

Hicks, L.A., Rose Jr., C.E., Fields, B.S., Drees, M.L., Engel, J.P., Jenkins, P.R., Rouse, B.S., Blythe, D., Khalifah, A.P., Feikin, D.R., Whitney, C.G., 2007. Increased rainfall is associated with increased risk for legionellosis. Epidemiol. Infect. 135, 811–817.

Holden, P.A., Fierer, N., 2005. Microbial processes in the vadose zone. Vadose Zone J. 4, 1–21.

IUSS, W.G.W., 2007. World Reference Base for Soil Resources 2006, First Update 2007. World Soil Resources Reports No. 103. FAO, Rome, Italy.

Jaesche, P., Totsche, K.U., Kögel-Knabner, I., 2006. Transport and anaerobic biodegradation of propylene glycol in gravel-rich soil materials. J. Contam. Hydrol. 85, 271–286.

Jiang, S., Pang, L., Buchan, G.D., Simunek, J., Noonan, M.J., Close, M.E., 2010. Modeling water flow and bacterial transport in undisturbed lysimeters under irrigations of dairy shed effluent and water using HYDRUS-1D. Water Res. 44, 1050–1061.

Joergensen, R.G., Wichern, F., 2008. Quantitative assessment of the fungal contribution to microbial tissue in soil. Soil Biol. Biochem. 40, 2977–2991.

Jones, R.T., Robeson, M.S., Lauber, C.L., Hamady, M., Knight, R., Fierer, N., 2009. A comprehensive survey of soil acidobacterial diversity using pyrosequencing and clone library analyses. ISME J. 3, 442–453.

Junier, P., Junier, T., Witzel, K.-P., 2008. TRiFLe, a program for in silico terminal restriction fragment length polymorphism analysis with user-defined sequence sets. Appl. Environ. Microbiol. 74, 6452–6456.

KA5, 2005. Bodenkundliche Kartieranleitung. Ad-Hoc-Arbeitsgruppe Boden: 5. verbesserte und erweiterte Auflage. E. Schweizerbart Science Publishers, ISBN 978-3-510-95920-4.

Kalbitz, K., Kaiser, K., 2008. Contribution of dissolved organic matter to carbon storage in forest mineral soils. J. Plant Nutr. Soil Sci. 171, 52–60.

Kieft, T.L., Murphy, E.M., Haldeman, D.L., Amy, P.S., Bjornstad, B.N., McDonald, E.V., Ringelberg, D.B., White, D.C., Stair, J., Griffiths, R.P., Gsell, T.C., Holben, W.E., Boone, D.R., 1998. Microbial transport, survival, and succession in a sequence of buried sediments. Microb. Ecol. 36, 336–348.

Kindler, R., Miltner, A., Thullner, M., Richnow, H.-H., Kaestner, M., 2009. Fate of bacterial biomass derived fatty acids in soil and their contribution to soil organic matter. Org. Geochem. 40, 29–37.

Kindler, R., Siemens, J., Kaiser, K., Walmsley, D.C., Bernhofer, C., Buchmann, N., Cellier, P., Eugster, W., Gleixner, G., Grunwald, T., Heim, A., Ibrom, A., Jones, S.K., Jones, M., Klumpp, K., Kutsch, W., Larsen, K.S., Lehuger, S., Loubet, B., McKenzie, R., Moors, E., Osborne, B., Pilegaard, K., Rebmann, C., Saunders, M., Schmidt, M.W.I., Schrumpf, M., Seyfferth, J., Skiba, U., Soussana, J.F., Sutton, M.A., Tefs, C., Vowinckel, B., Zeeman, M.J., Kaupenjohann, M., 2011. Dissolved carbon leaching from soil is a crucial component of the net ecosystem carbon balance. Glob. Change Biol. 17, 1167–1185.

Knox, O.G.G., Killham, K., Artz, R.R.E., Mullins, C., Wilson, M., 2004. Effect of nematodes on rhizosphere colonization by seed-applied bacteria. Appl. Environ. Microbiol. 70, 4666–4671.

Kögel-Knabner, I., 2002. The macromolecular organic composition of plant and microbial residues as inputs to soil organic matter. Soil Biol. Biochem. 34, 139–162.

Kramer, S., Marhan, S., Ruess, L., Armbruster, W., Butenschoen, O., Haslwimmer, H., Kuzyakov, Y., Pausch, J., Scheunemann, N., Schoene, J., Schmalwasser, A., Totsche, K.U., Walker, F., Scheu, S., Kandeler, E., 2012. Carbon flow into microbial and fungal biomass as a basis for the belowground food web of agroecosystems. Pedobiologia 55, 111–119.

Le, S., Josse, J., Husson, F., 2008. FactoMineR: an R package for multivariate analysis. J. Stat. Softw. 25, 1–18.

Lee, C.K., Herbold, C.W., Polson, S.W., Wommack, K.E., Williamson, S.J., McDonald, I.R., Cary, S.C., 2012. Groundtruthing next-gen sequencing for microbial ecology–biases and errors in community structure estimates from PCR amplicon pyrosequencing. PLoS One 7, e44224.

Liang, C., Balser, T.C., 2011. Microbial production of recalcitrant organic matter in global soils: implications for productivity and climate policy. Nat. Rev. Microbiol. 9, 75.

Litton, C.M., Giardina, C.P., 2008. Below-ground carbon flux and partitioning: global patterns and response to temperature. Funct. Ecol. 22, 941–954.

Loferer-Krossbacher, M., Klima, J., Psenner, R., 1998. Determination of bacterial cell dry mass by transmission electron microscopy and densitometric image analysis. Appl. Environ. Microbiol. 64, 688–694.

Lueders, T., Manefield, M., Friedrich, M.W., 2004. Enhanced sensitivity of DNA- and rRNA-based stable isotope probing by fractionation and quantitative analysis of isopycnic centrifugation gradients. Environ. Microbiol. 6, 73–78.

Markowitz, V.M., Chen, I.-M.A., Palaniappan, K., Chu, K., Szeto, E., Grechkin, Y., Ratner, A., Jacob, B., Huang, J., Williams, P., Huntemann, M., Anderson, I., Mavromatis, K., Ivanova, N.N., Kyrpides, N.C., 2012. IMG: the integrated microbial genomes database and comparative analysis system. Nucleic Acids Res. 40, D115–D122.

Martins, J.M.F., Majdalani, S., Vitorge, E., Desaunay, A., Navel, A., Guine, V., Daian, J.F., Vince, E., Denis, H., Gaudet, J.P., 2013. Role of macropore flow in the transport of *Escherichia coli* cells in undisturbed cores of a brown leached soil. Environ. Sci. Process. Impacts 15, 347–356.

Miltner, A., Bombach, P., Schmidt-Brücken, B., Kästner, M., 2012. SOM genesis: microbial biomass as a significant source. Biogeochemistry 111, 41–55.

Natsch, A., Keel, C., Troxler, J., Zala, M., VonAlbertini, N., Defago, G., 1996. Importance of preferential flow and soil management in vertical transport of a biocontrol strain of Pseudomonas fluorescens in structured field soil. Appl. Environ. Microbiol. 62, 33–40.

Pilloni, G., Granitsiotis, M.S., Engel, M., Lueders, T., 2012. Testing the limits of 454 pyrotag sequencing: reproducibility, quantitative assessment and comparison to T-RFLP fingerprinting of aquifer microbes. PLoS One 7, e40467.

Pilloni, G., von Netzer, F., Engel, M., Lueders, T., 2011. Electron acceptor-dependent identification of key anaerobic toluene degraders at a tar-oil-contaminated aquifer by Pyro-SIP. Fems Microbiol. Ecol. 78, 165–175.

R Development Core Team, 2011. A language and environment for statistical computing. In: Computing, R.F.F.S. (Ed.), 2.14.0 ed. R Foundation for Statistical Computing, Vienna, Austria.

Scharroba, A., Dibbern, D., Hünninghaus, M., Kramer, S., Moll, J., Butenschoen, O., Bonkowski, M., Buscot, F., Kandeler, E., Koller, R., Krüger, D., Lueders, T., Scheu, S., Ruess, L., 2012. Effects of resource availability and quality on the structure of the micro-food web of an arable soil across depth. Soil Biol. Biochem. 50, 1–11.

Schütte, U.M.E., Abdo, Z., Bent, S.J., Shyu, C., Williams, C.J., Pierson, J.D., Forney, L.J., 2008. Advances in the use of terminal restriction fragment length polymorphism (T-RFLP) analysis of 16S rRNA genes to characterize microbial communities. Appl. Microbiol. Biotechnol. 80, 365–380.

Sen, T.K., 2011. Processes in pathogenic biocolloidal contaminants transport in saturated and unsaturated porous media: a review. Water Air Soil Pollut. 216, 239–256.

Smith, C.J., Danilowicz, B.S., Clear, A.K., Costello, F.J., Wilson, B., Meijer, W.G., 2005. T-Align, a web-based tool for comparison of multiple terminal restriction fragment length polymorphism profiles. Fems Microbiol. Ecol. 54, 375–380.

Totsche, K.U., Jann, S., Kögel-Knabner, I., 2007. Single event-driven export of polycyclic aromatic hydrocarbons and suspended matter from coal tar-contaminated soil. Vadose Zone J. 6, 233–243.
Troussellier, M., Bouvy, M., Courties, C., Dupuy, C., 1997. Variation of carbon content among bacterial species under starvation condition. Aquat. Microb. Ecol. 13, 113–119.
Troxler, J., Zala, M., Natsch, A., Nievergelt, J., Keel, C., Defago, G., 1998. Transport of a biocontrol Pseudomonas fluorescens through 2.5-m deep outdoor lysimeters and survival in the effluent water. Soil Biol. Biochem. 30, 621–631.
Tuomi, P., Fagerbakke, K.M., Bratbak, G., Heldal, M., 1995. Nutritional enrichment of a microbial community: the effects on activity, elemental composition, community structure and virus production. FEMS Microbiol. Ecol. 16, 123–134.
Unc, A., Goss, M.J., 2004. Transport of bacteria from manure and protection of water resources. Appl. Soil Ecol. 25, 1–18.
Vance, E.D., Brookes, P.C., Jenkinson, D.S., 1987. An extraction method for measuring soil microbial biomass carbon. Soil Biol. Biochem. 19, 703–708.
Vinther, F.P., Eiland, F., Lind, A.M., Elsgaard, L., 1999. Microbial biomass and numbers of denitrifiers related to macropore channels in agricultural and forest soils. Soil Biol. Biochem. 31, 603–611.
Wang, Q., Garrity, G.M., Tiedje, J.M., Cole, J.R., 2007. Naive Bayesian classifier for rapid assignment of rRNA sequences into the new bacterial taxonomy. Appl. Environ. Microbiol. 73, 5261–5267.
Weissbrodt, D.G., Shani, N., Sinclair, L., Lefebvre, G., Rossi, P., Maillard, J., Rougemont, J., Holliger, C., 2012. PyroTRF-ID: a novel bioinformatics methodology for the affiliation of terminal-restriction fragments using 16S rRNA gene pyrosequencing data. BMC Microbiol. 12, 306.
Will, C., Thurmer, A., Wollherr, A., Nacke, H., Herold, N., Schrumpf, M., Gutknecht, J., Wubet, T., Buscot, F., Daniel, R., 2010. Horizon-specific bacterial community composition of German grassland soils, as revealed by pyrosequencing-based analysis of 16S rRNA genes. Appl. Environ. Microbiol. 76, 6751–6759.
Winderl, C., Anneser, B., Griebler, C., Meckenstock, R.U., Lueders, T., 2008. Depth-resolved quantification of anaerobic toluene degraders and aquifer microbial community patterns in distinct redox zones of a tar oil contaminant plume. Appl. Environ. Microbiol. 74, 792–801.